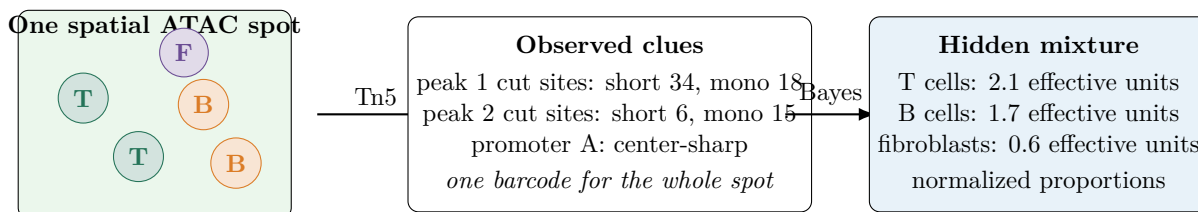


ShapeMix-ATAC

A LaTeX tutorial for the Bayesian fragment-shape model

Written to make the equations, parameters, priors, and hyperparameters easier to read



Main idea

A spatial ATAC spot is treated as a mixture of cell-type chromatin fingerprints. The primary experiment asks whether the parent-fragment length composition of deduplicated Tn5 cut sites improves deconvolution beyond their total counts in peaks.

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1 What problem is ShapeMix-ATAC solving?

Spatial ATAC-seq measures open chromatin in a tissue section. Its fragments file records deduplicated DNA fragments assigned to spatial barcodes. Each fragment provides two Tn5 cut sites, and a spot may contain more than one nucleus. Therefore, the observed signal is a mixture.

Biology intuition

Think of a spot as a small square drawn on top of tissue. Inside that square, there may be T cells, B cells, fibroblasts, tumor cells, neurons, or other cell types. ATAC-seq does not directly say which nucleus produced each fragment or cut site. ShapeMix-ATAC tries to infer the hidden cell mixture from the cut-site counts and their parent fragments' lengths.

The ordinary deconvolution question is:

$$\text{observed mixed signal} \implies \text{cell-type amounts in the spot.}$$

The ShapeMix-ATAC question is richer:

$$\begin{array}{ccc} \text{observed peak totals and} & & \text{effective abundances and} \\ \text{parent-fragment length composition} & \implies & \text{normalized cell-type proportions} \end{array}$$

1.1 Why ATAC needs a special model

Spatial transcriptomics mostly counts RNA molecules from genes. Standard Cell Ranger ATAC peak matrices count Tn5 cut sites in open-chromatin peaks. The parent fragments also have physical features:

- **Fragment length:** short fragments often come from nucleosome-free DNA; longer fragments can span nucleosomes.
- **Position inside a peak:** fragments may concentrate near the center, left edge, right edge, or flanking regions.
- **Motif-centered pattern:** cuts around a transcription-factor motif can carry footprint-like information.

Main idea

The primary ShapeMix data object is a three-dimensional count table:

$$Y_{s,p,b} = \text{deduplicated Tn5 cut sites in spot } s, \text{ peak } p, \text{ parent-length bin } b.$$

Each cut site inherits the length bin of its parent fragment. A usual Cell Ranger-compatible peak-count model uses only $N_{s,p} = \sum_b Y_{s,p,b}$. ShapeMix keeps the extra index b . A unique-fragment-count analysis would be a separately versioned sensitivity analysis with its own matched baseline.

2 The data table: from peak counts to shape counts

2.1 Indices

Symbol	Meaning
$s = 1, \dots, S$	spatial spots or pixels
$p = 1, \dots, P$	ATAC peaks or peak groups
$b = 1, \dots, B$	bins of parent-fragment length for observed cut sites
$c = 1, \dots, C$	cell types in the reference, such as T cell, B cell, fibroblast

2.2 Shape-bin examples

A first implementation should use only a few bins. This keeps the method explainable and avoids overfitting.

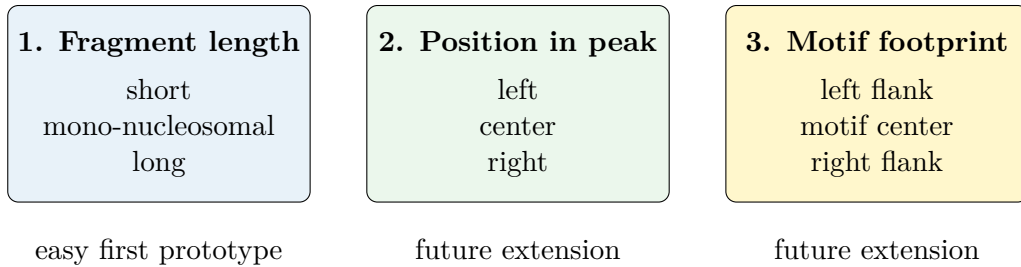


Figure 1: Three possible choices for the shape-bin index b . A first prototype should usually start with fragment-length bins.

The primary implementation uses exactly three parent-fragment length bins:

$$b = 1 : [0, 100), \quad b = 2 : [100, 250), \quad b = 3 : [250, \infty) \text{ bp.}$$

Length is `chromEnd` – `chromStart` from the adjusted, zero-based half-open 10x fragment interval. Each fragments-file row is already a deduplicated fragment; the `readSupport` field is ignored in the primary analysis so PCR read-pair support is not counted again. Both cut sites inherit the same parent-length bin and are assigned independently to the unique selected, non-overlapping peak containing them. The right-cut coordinate convention must be validated by reconstructing selected columns of the official peak matrix before canonical data are frozen.

3 A short Bayesian warm-up

Bayesian modeling means we write down a story for how the data could have been generated. Then we run the story backward.

Bayes rule in words

$$\text{posterior} \propto \text{likelihood} \times \text{prior}.$$

In ShapeMix-ATAC:

$$\begin{aligned}
& p(\text{hidden cell mixture} \mid \text{observed ATAC cut-site tensor}) \\
& \propto p(\text{observed ATAC cut-site tensor} \mid \text{hidden cell mixture}) \\
& \quad \times p(\text{hidden cell mixture}).
\end{aligned}$$

3.1 Distribution cheat sheet

Distribution	Used for	Intuition
Poisson	simple counts	Counts fluctuate around their mean.
Negative binomial	noisy sequencing counts	Like Poisson, but allows extra noise.
Gamma	positive amounts	Good for abundances, background, dispersion, and rates.
Log-normal	positive multiplicative effects	Good for depth/yield/bias factors.
Beta	probabilities between 0 and 1	Good for peak accessibility probabilities.
Dirichlet	vectors that sum to 1	Good for proportions, such as shape-bin probabilities.
Multinomial	splitting a fixed peak total among bins	Primary conditional likelihood for parent-length composition.
Normal / CAR normal	real-valued latent spatial fields	Good for smooth spatial variation.

4 Start with the ordinary peak-count model

The primary model fixes reference signatures and infers only spot abundances. Let $R_{i,p,b}$ be the training-reference cut-site count for cell i , peak p , and parent-length bin b , and define $T_{i,p} = \sum_b R_{i,p,b}$. From these training cells we estimate $A_{c,p}$, the mean total cut-site rate per reference cell of type c , $\omega_{c,p,b}$, its conditional parent-length-bin distribution, and one global per-reference-cell inverse-dispersion ϕ_{ref} .

Let $z_{s,c} > 0$ be the unknown effective abundance and $e_s = \sum_c z_{s,c}$. The ordinary peak total and its expected value are

$$N_{s,p} = \sum_b Y_{s,p,b}, \quad (1)$$

$$\mu_{s,p} = \sum_c z_{s,c} A_{c,p}. \quad (2)$$

The count-only model is

Shared total-count model

$$N_{s,p} \sim \text{NegBinomial}(\text{mean} = \mu_{s,p}, \text{dispersion} = \max(e_s, \epsilon) \phi_{\text{ref}}). \quad (3)$$

There is no observed spot-depth offset in the primary model. Total captured depth is absorbed into z .

5 The ShapeMix-ATAC likelihood

5.1 The factorized core equation

ShapeMix preserves Equation (3) and adds a conditional allocation of the same total among parent-length bins. Define

$$v_{s,p,b} = \sum_c z_{s,c} A_{c,p} \omega_{c,p,b}, \quad (4)$$

$$\mu_{s,p} = \sum_b v_{s,p,b} = \sum_c z_{s,c} A_{c,p}, \quad (5)$$

$$\rho_{s,p,b} = \frac{v_{s,p,b}}{\mu_{s,p}}. \quad (6)$$

The primary same-dataset simulation has zero background. Its complete likelihood is:

Core ShapeMix likelihood

$$N_{s,p} \sim \text{NegBinomial}(\text{mean} = \mu_{s,p}, \text{dispersion} = \max(e_s, \epsilon) \phi_{\text{ref}}), \quad (7)$$

$$\mathbf{Y}_{s,p,\cdot} \mid N_{s,p} \sim \text{Multinomial}(N_{s,p}, \boldsymbol{\rho}_{s,p,\cdot}). \quad (8)$$

The first line asks how many cut sites occur in the peak. The second asks how that fixed total is distributed among the three parent-fragment length bins. The new ShapeMix information is $\omega_{c,p,b}$, the reference conditional shape.

Important caution

Independent negative-binomial likelihoods for $Y_{s,p,b}$ are not the primary model. Their sum is generally not the same negative-binomial total as the count-only arm. They remain only as a clearly named future sensitivity model.

5.2 Negative-binomial parameterization

This document uses a negative binomial with mean μ and inverse-dispersion ϕ :

$$X \sim \text{NegBinomial}(\text{mean} = \mu, \text{dispersion} = \phi)$$

means

$$\mathbb{E}[X] = \mu, \quad \text{Var}(X) = \mu + \frac{\mu^2}{\phi}.$$

A spot uses $\phi = \max(e_s, \epsilon) \phi_{\text{ref}}$. Scaling by effective abundance approximates aggregation of independent reference-depth cells. The conditional log likelihood is evaluated stably as

$$\log p(\mathbf{Y} \mid N, \boldsymbol{\rho}) = \log \Gamma(N+1) - \sum_b \log \Gamma(Y_b+1) + \sum_{b: Y_b > 0} Y_b \log \rho_b.$$

Rows with $N = 0$ contribute exactly zero. In the Poisson-total special case, this factorization is algebraically equivalent to independent Poisson bin counts; that identity is an implementation test.

5.3 The exact count-only ablation

The two objectives are

$$\begin{aligned} \mathcal{L}_{\text{count}}(z) &= \log p(N \mid z, A, \phi_{\text{ref}}) + \log p(z), \\ \mathcal{L}_{\text{shape}}(z) &= \mathcal{L}_{\text{count}}(z) + \log p(\mathbf{Y} \mid N, z, A, \omega). \end{aligned}$$

Everything except the final conditional shape term is identical: outer split, inner mixture, peaks, N , A , ϕ_{ref} , abundance prior, parameters, initialization, optimizer, stopping rule, seed, and compute budget. If all cell types have identical $\omega_{c,p,\cdot}$, the shape term contains no information about z , so both arms should agree within numerical tolerance.

6 The generative story

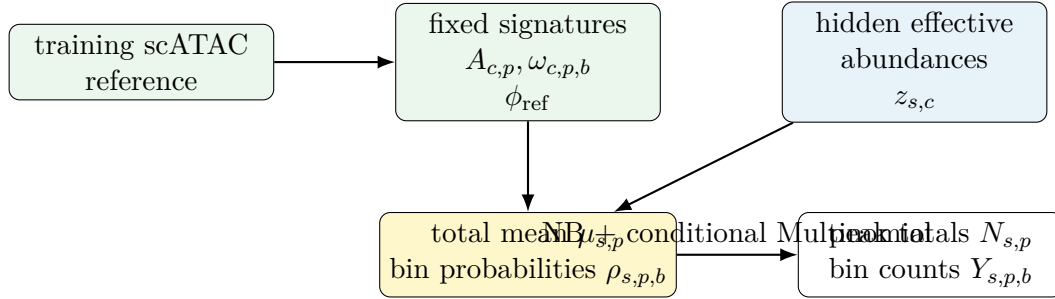


Figure 2: The primary ShapeMix model keeps signatures and dispersion fixed and infers only positive abundances.

The model can be taught as a recipe:

1. Use only training-reference cells to select peaks and estimate A , ω , and ϕ_{ref} .
2. Form pseudo-spots only from held-out test cells.
3. Infer positive $z_{s,c}$ from the shared total-count likelihood.
4. For ShapeMix only, add the conditional parent-length composition likelihood.
5. Normalize z to obtain cell-type proportions and evaluate paired runs.

7 Reference signatures

The primary model treats signatures as empirical inputs, not inferred latent variables. This avoids trying to decompose the same scale simultaneously among abundance, cell-type yield, peak detectability, and accessibility.

Let $\mathcal{I}_c$ be training-reference cells of type c , g_p a pooled peak-rate target, $u_{p,b}$ the pooled training-reference parent-length distribution for peak p , and $h_i = 1$ for the primary same-protocol experiment. Define a dataset-global distribution and use the same frozen $\alpha_\omega = 1$ to smooth the pooled peak distribution toward it:

$$u_b^{\text{global}} = \frac{\sum_{i,p} R_{i,p,b}}{\sum_{i,p,b'} R_{i,p,b'}}, \quad (9)$$

$$u_{p,b} = \frac{\sum_i R_{i,p,b} + \alpha_\omega u_b^{\text{global}}}{\sum_{i,b'} R_{i,p,b'} + \alpha_\omega}. \quad (10)$$

Every global bin must have positive training-reference support. Cell-type signatures then use

$$A_{c,p} = \frac{\sum_{i \in \mathcal{I}_c} T_{i,p} + \alpha_A g_p}{\sum_{i \in \mathcal{I}_c} h_i + \alpha_A}, \quad (11)$$

$$\omega_{c,p,b} = \frac{\sum_{i \in \mathcal{I}_c} R_{i,p,b} + \alpha_\omega u_{p,b}}{\sum_{i \in \mathcal{I}_c} T_{i,p} + \alpha_\omega}. \quad (12)$$

Sparse peaks shrink first toward their pooled peak distribution and then toward the dataset-global distribution. The concentration is frozen by the benchmark protocol and is not tuned on held-out mixtures.

Biology intuition

If T-cell cut sites at peak p usually have short parent fragments, then $\omega_{T,p,\text{short}}$ is high. If B-cell cut sites at the same peak mostly have mononucleosomal-length parents, ShapeMix can use that conditional difference even when $A_{T,p}$ and $A_{B,p}$ are similar.

7.1 Cross-fitted global dispersion

Inside the training-reference pool, make a deterministic, cell-type-stratified two-fold split. For each cell i , estimate its held-out mean $m_{i,p}$ from the opposite fold's cell-type signature. Then compute

$$\alpha_{\text{ref}}^{\text{raw}} = \frac{\sum_{i,p} [(T_{i,p} - m_{i,p})^2 - m_{i,p}]}{\sum_{i,p} m_{i,p}^2}, \quad (13)$$

$$\alpha_{\text{ref}} = \max(\alpha_{\text{ref}}^{\text{raw}}, 10^{-8}), \quad (14)$$

$$\phi_{\text{ref}} = 1/\alpha_{\text{ref}}. \quad (15)$$

The fold seed, numerator, denominator, and resulting values are recorded. A non-finite or zero denominator is an error. After cross-fitting, A is re-estimated from all training-reference cells. Peak-specific or learned dispersion is a future sensitivity analysis.

8 Prior and semantics for hidden abundance

The only inferred biological quantity in the primary MAP model is $z_{s,c} > 0$. A simple fixed Gamma prior is

$$z_{s,c} \sim \text{Gamma}\left(r_z, \frac{r_z}{m_c}\right), \quad \mathbb{E}[z_{s,c}] = m_c. \quad (16)$$

The implementation uses $z = \text{softplus}(z^{\text{raw}}) + \epsilon$ and adds the corresponding log prior to the objective.

Important caution

The primary model has no observed library-size offset ℓ_s . Because total captured depth is absorbed into z , $z_{s,c}$ is an effective reference-cell-equivalent, depth-scaled abundance. It is not guaranteed to be an integer or a calibrated nucleus count. The standardized scientific output is

$$\pi_{s,c} = \frac{z_{s,c}}{\sum_{c'} z_{s,c'}}.$$

9 Complete core model

Putting the primary pieces together:

Complete core ShapeMix-ATAC model

$$v_{s,p,b} = \sum_c z_{s,c} A_{c,p} \omega_{c,p,b}, \quad (17)$$

$$\mu_{s,p} = \sum_b v_{s,p,b}, \quad \rho_{s,p,b} = v_{s,p,b} / \mu_{s,p}, \quad (18)$$

$$N_{s,p} \sim \text{NegBinomial}(\mu_{s,p}, \max(e_s, \epsilon) \phi_{\text{ref}}), \quad (19)$$

$$\mathbf{Y}_{s,p,\cdot} \mid N_{s,p} \sim \text{Multinomial}(N_{s,p}, \boldsymbol{\rho}_{s,p,\cdot}), \quad (20)$$

$$z_{s,c} \sim \text{Gamma}(r_z, r_z / m_c), \quad e_s = \sum_c z_{s,c}. \quad (21)$$

The fixed quantities A , ω , and ϕ_{ref} come only from training-reference cells. The output cell-type proportion is computed from MAP abundances:

$$\pi_{s,c} = \frac{z_{s,c}}{\sum_{c'=1}^C z_{s,c'}}.$$

10 Primary parameter guide

Table 1: Quantities in the executable MVP.

Symbol	Meaning	Status
$Y_{s,p,b}$	deduplicated cut-site count annotated by parent-length bin	observed sparse layer
$N_{s,p}$	$\sum_b Y_{s,p,b}$, the ordinary peak total	observed; identical in both arms
$A_{c,p}$	mean total cut-site rate per training-reference cell	fixed smoothed estimate
$\omega_{c,p,b}$	conditional parent-length-bin probability	fixed smoothed estimate
ϕ_{ref}	global per-reference-cell NB inverse-dispersion	fixed cross-fitted estimate
$z_{s,c}$	effective reference-cell-equivalent, depth-scaled abundance	inferred by MAP
e_s	$\sum_c z_{s,c}$, effective abundance scale	deterministic
$\mu_{s,p}$	expected total cut-site count	deterministic
$\rho_{s,p,b}$	expected conditional bin probability	deterministic
$\pi_{s,c}$	row-normalized cell-type proportion	reported output
$r_z, m_c, \alpha_A, \alpha_\omega, \epsilon$	prior, smoothing, and numerical settings	fixed before final evaluation

11 Richer Bayesian components are future extensions

The earlier model draft included a learned background $\beta_{p,b}$, cell-type yield q_c , peak detectability d_p , latent accessibility $\theta_{c,p}$, inferred ω , spatial effects $u_{s,c}$, peak-specific dispersion ϕ_p , and variational uncertainty.

These remain useful research directions, but none belongs to the primary ablation. Estimating all of them with z would create scale and attribution ambiguities and could let background or spatial smoothing absorb the shape signal.

Extensions must be introduced one at a time after the three-bin MAP model passes its recovery, conservation, and negative-control tests. A fixed small background is the first robustness check; a Dirichlet–multinomial conditional shape model is the first shape-overdispersion check. Learned background, cross-protocol scaling, spatial smoothing, richer position or motif axes, learned signatures, and variational inference follow only when residual diagnostics justify them. The independent-bin negative-binomial formulation is a sensitivity model, never the count-only control for the primary scientific claim.

12 Worked example: why shape helps

Suppose there are only two cell types: T cells and B cells. At one peak, both cell types have the same total accessibility, so a peak-count-only model cannot distinguish them.

Cell type / spot	Total cut sites at peak	Short-parent cut sites	Mono-parent cut sites
T-cell reference	100	80%	20%
B-cell reference	100	20%	80%
Observed spot	100	60%	40%

Peak-count model sees only:

$$100 \approx 100.$$

It learns almost nothing.

ShapeMix sees:

$$0.60 \approx 0.80\pi_T + 0.20(1 - \pi_T).$$

Solving:

$$0.60 = 0.80\pi_T + 0.20 - 0.20\pi_T,$$

$$0.40 = 0.60\pi_T,$$

$$\pi_T \approx 0.67.$$

So the observed spot looks like roughly:

67% T cell	+	33% B cell.
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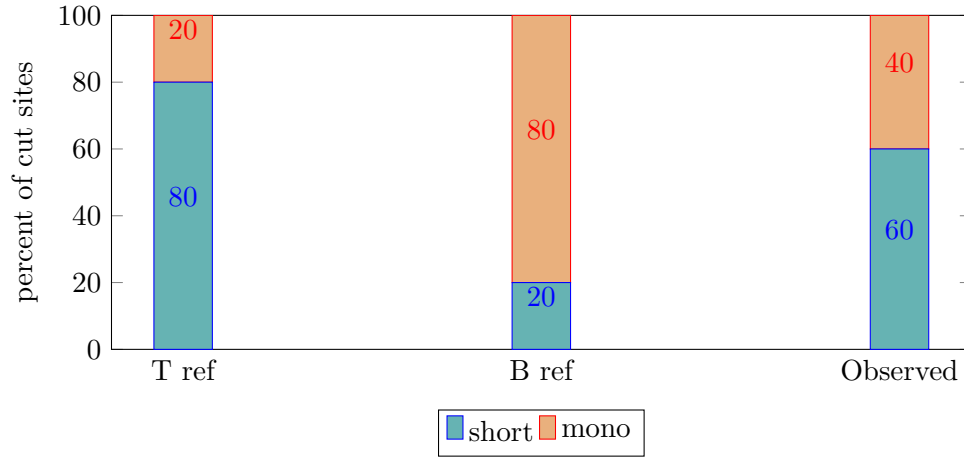


Figure 3: The total peak count is identical, but the shape distribution reveals a mixture.

13 What the primary MAP estimate means

For fixed A , ω , and ϕ_{ref} , the primary model finds

$$\hat{z} = \arg \max_{z \geq 0} \{ \log p(N \mid z, A, \phi_{\text{ref}}) + \mathbb{K}_{\text{shape}} \log p(Y \mid N, z, A, \omega) + \log p(z) \}.$$

The reported estimate is $\hat{\pi}_{s,c} = \hat{z}_{s,c} / \sum_{c'} \hat{z}_{s,c'}$. It is a point estimate. The MAP MVP does not produce credible intervals, and its outputs must not be described as posterior means or calibrated uncertainty.

Future uncertainty model

Uncertainty remains scientifically valuable, especially for rare cell types. Variational inference may be added after the MAP likelihood, gradients, recovery, and negative controls are validated. That extension must be reported as a new model version rather than attributed to the MVP.

14 Fitting the model

14.1 Primary: MAP estimation

MAP means maximum a posteriori. It finds the single parameter setting that maximizes:

$$\log p(N \mid z) + \mathbb{K}_{\text{shape}} \log p(Y \mid N, z) + \log p(z),$$

with only z inferred. Initialize from collapsed-count NNLS with a uniform fallback, use deterministic restarts, optimize with Adam, and fail on non-finite losses, gradients, abundances, or proportions. Process peaks and spots in chunks rather than materializing a dense $S \times P \times B$ expected tensor.

14.2 Future: variational inference

Variational inference approximates the posterior with a simpler distribution:

$$q(z) \approx p(z \mid Y),$$

Then it optimizes an evidence lower bound, usually called the ELBO:

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_q[\log p(Y, \Theta)] - \mathbb{E}_q[\log q(\Theta)].$$

This can provide approximate uncertainty, but it is not part of the primary experiment and should not be implemented until MAP behavior is credible.

15 Leakage-free experimental structure

The available PBMC input contains one donor. Therefore the primary experiment uses deterministic, cell-type-stratified 70/30 cell splits rather than claiming a donor split. Each *outer split seed* defines one reference/test partition. Peak selection, smoothing, signatures, and dispersion use only the 70% reference pool. Each *inner mixture seed* samples pseudo-spots only from that outer split's held-out 30% pool.

Both arms run on the exact same pseudo-spots for every (outer seed, inner seed, condition) tuple. Inner-mixture effects are first averaged within each outer split before describing variability across outer splits. Neither spots nor inner mixtures are independent biological replicates. Intervals from this design quantify conditional resampling variability within one donor and do not establish donor-level generalization.

16 How to build the shape-count tensor

16.1 Step-by-step preprocessing

1. Start with the BGZF/tabix-indexed five-column 10x fragments file: chromosome, start, end, barcode, and **readSupport**.
2. Treat each row as one deduplicated fragment and ignore **readSupport** in the primary analysis.
3. Compute parent length $L = \text{chromEnd} - \text{chromStart}$ and assign both of its Tn5 cut sites to one fixed length bin.
4. Assign each cut site to the unique selected, non-overlapping Cell Ranger peak that contains it. Validate the right-cut coordinate against the vendor peak matrix first.
5. Select 5,000 peaks by a deterministic accessibility-variance rank computed from reference cells only; never use held-out cells or shape counts for selection.
6. Accumulate sparse cell/spot-by-peak matrices for each bin and set $N_{s,p} = \sum_b Y_{s,p,b}$.
7. Require the layer sum to equal the ordinary peak-count matrix exactly and record source, feature, and split hashes.

16.2 Fragment-length bin example

For fragment length L :

$$b(L) = \begin{cases} 1, & L < 100 \quad \text{short,} \\ 2, & 100 \leq L < 250 \quad \text{mono-nucleosomal-like,} \\ 3, & L \geq 250 \quad \text{long.} \end{cases}$$

These half-open cutoffs are fixed for the primary experiment. Alternative cutoffs are versioned sensitivity analyses, not tuning on final mixtures.

17 Diagnostics and ablation tests

17.1 Diagnostics

After fitting, check:

- **Conservation:** bin layers sum exactly to N , and selected columns reconstruct or explain the official vendor matrix.
- **Count reconstruction:** compare fitted μ with N by peak and spot.
- **Shape-bin reconstruction:** does the model reproduce short/mono/long ratios?
- **Optimization:** all losses, gradients, abundances, and proportions are finite; restarts agree or their disagreement is reported.
- **Homogenized shape:** identical ω across cell types makes ShapeMix agree with count-only within tolerance.
- **Permuted shape:** permuting cell-type shape labels must not create a credible improvement.
- **Poisson identity:** factorized Poisson total plus conditional multinomial equals independent Poisson bin counts.

17.2 Ablation tests

Ablation means removing one ingredient at a time to test whether it matters.

Model version	Question answered
Shared NB total only	Primary count-only arm.
Shared NB total + conditional length composition	Primary ShapeMix arm; only the conditional term differs.
Cell-type-homogenized ω	Negative control: does shape become non-informative as required?
Cell-type-permuted ω	Negative control: can arbitrary shape labels appear beneficial?
Poisson versus NB total	Sensitivity to total-count overdispersion.
Zero versus fixed small background	Robustness to unexplained signal.
One/three bins and alternative cutoffs	Sensitivity to binning, using versioned datasets.
5,000/10,000/20,000 peaks	Sensitivity to reference-only feature count.

18 What makes ShapeMix-ATAC new and testable?

The standard transferred deconvolution methods use spot-by-feature count matrices. For ATAC, the feature is usually a peak. ShapeMix augments that matrix with a tensor whose entries use the same count unit:

$$\text{ordinary ATAC deconvolution: } N_{s,p} \quad \Longrightarrow \quad \text{ShapeMix-ATAC: } Y_{s,p,b}.$$

Main idea

The narrow scientific claim is: *does the parent-fragment length composition of observed insertions improve deconvolution beyond their peak totals?*

This is especially useful for:

- closely related immune subtypes,
- nearby developmental states,
- tumor and stromal mixtures,
- rare cell types with subtle peak-count differences,
- cell types that share open peaks but differ in parent-fragment length patterns.

19 Implementation pseudocode

Pseudocode

Input:

BGZF fragments file plus tabix index
 non-overlapping Cell Ranger peaks
 scATAC/multiome reference with cell type labels

Preprocess:

make deterministic stratified 70/30 outer cell split
 choose 5,000 peaks from reference cells only
 ignore readSupport; bin each deduplicated parent fragment
 assign its two cut sites to containing peaks
 build sparse layers $Y[s,p,b]$ and totals $N[s,p]$
 estimate fixed $A[c,p]$, $\omega[c,p,b]$, ϕ_{ref} from reference

For each nested mixture seed and each arm:

form the same pseudo-spots from held-out cells
 initialize $z[s,c]$ from collapsed-count NNLS
 repeat until convergence:
 compute $\mu[s,p]$ and abundance-scaled NB total loss
 if shape arm: add conditional multinomial loss
 update only positive $z[s,c]$ plus its fixed prior

Output:

$\pi[s,c] = z[s,c] / \sum_c z[s,c]$
 effective depth-scaled abundance $z[s,c]$
 convergence and restart diagnostics
 residual peaks and residual shape patterns

20 Common questions

20.1 Is $\omega_{c,p,b}$ inferred from each spot?

No. In the primary model, ω is a fixed, smoothed empirical input estimated only from training-reference cells. Inferring it jointly with abundance is a future model version.

20.2 Why is there no $\theta_{c,p}$, yield, or detectability term?

Their product is absorbed into the fixed total-rate signature $A_{c,p}$. This keeps the MVP identifiable enough to test shape without asking the model to divide one scale among z , yield, detectability, and accessibility.

20.3 Is $z_{s,c}$ a cell count?

No. It is an effective reference-cell-equivalent, depth-scaled abundance. It may track cell number in a same-protocol simulation, but without an observed depth offset or external calibration it is neither an exact integer nor a nucleus count. The primary biological output is the normalized proportion π .

20.4 Can we use Poisson instead of negative binomial?

Yes. Replace only the shared total model with $N_{s,p} \sim \text{Poisson}(\mu_{s,p})$ and retain the same conditional multinomial. This is also a golden test because the resulting joint distribution equals independent Poisson bin counts with means $v_{s,p,b}$.

20.5 Do we need all peaks?

No. The primary run uses 5,000 peaks selected by a frozen reference-only accessibility-variance ranking. Runs with 10,000 and 20,000 peaks are sensitivity analyses. Selecting peaks from held-out mixtures or from shape differences would leak test information into the model.

21 Fixed evaluation contract

The dataset declares one ordered cell-type universe, which must equal the truth columns and reference-label universe. Prediction rows must contain the exact truth spot set. A missing declared prediction type is inserted as zero so omission is penalized; an unknown extra type is an error. Duplicate labels, non-numeric, non-finite, negative, or all-zero rows fail evaluation. Truth and prediction rows must already sum to one within the frozen tolerance; they are not silently renormalized and invalid rows are never dropped.

The primary metrics are:

- **RMSE:** root mean squared error across all $S \times C$ entries in the fixed universe.
- **jsd_v2:** mean per-spot base-2 Jensen–Shannon *divergence*,

$$\frac{1}{S} \sum_s \text{JSD}_2(\pi_s^{\text{true}}, \hat{\pi}_s),$$

operationally `mean(jensenshannon(truth, prediction, base=2)`²). Its range is $[0, 1]$. The square is required because the library function returns Jensen–Shannon distance; the historical unsquared value is a different metric.

Secondary metrics include per-type MAE and RMSE, correlation, frozen rare-cell precision, recall, F1, and AUPRC, plus runtime and memory. The metric version and declared universe are saved with every comparison.

22 Optional extensions after the primary test

Image-derived nuclei budgets, peak reliability weights, a fixed background, cross-protocol scaling, a Dirichlet–multinomial shape term, spatial smoothing, position or motif axes, Bayesian signatures, and variational uncertainty may all be valuable. Each changes the model or claim and therefore requires a versioned dataset/configuration, its own negative control, and an identifiability check. None is part of the first ShapeMix-versus-count-only comparison.

23 Recommended first experiment

1. Make frozen, stratified 70/30 outer splits and document the one-donor limitation.
2. From each reference pool only, choose 5,000 peaks and estimate A , ω , and cross-fitted ϕ_{ref} .
3. Build matched pseudo-spots from held-out cells with mixture seeds nested inside each split seed.
4. Fit the shared negative-binomial total model to both arms; add only the conditional three-bin term for ShapeMix.
5. Pair results within every outer-seed, inner-seed, and condition tuple. Average inner effects within outer splits before summarizing across splits.
6. Report RMSE and `jsd_v2`, plus frozen secondary metrics, runtime, convergence, and negative controls, even if shape does not help.

Important caution

Do not treat spots, inner mixtures, or repeated partitions of one donor as independent biological replicates. The primary result is evidence conditional on this dataset; cross-donor validation requires additional donors.

24 Summary in one page

The whole primary model in two lines

$$N_{s,p} \sim \text{NegBinomial}\left(\mu_{s,p} = \sum_c z_{s,c} A_{c,p}, \max\left(\sum_c z_{s,c}, \epsilon\right) \phi_{\text{ref}}\right),$$

$$\mathbf{Y}_{s,p,\cdot} \mid N_{s,p} \sim \text{Multinomial}\left(N_{s,p}, \frac{\sum_c z_{s,c} A_{c,p} \boldsymbol{\omega}_{c,p,\cdot}}{\sum_c z_{s,c} A_{c,p}}\right).$$

- $Y_{s,p,b}$: deduplicated cut sites in a peak, grouped by their parent fragment’s fixed length bin.
- $N_{s,p} = \sum_b Y_{s,p,b}$: the identical peak total used by both arms.
- $A_{c,p}$: fixed reference total-rate signature.
- $\omega_{c,p,b}$: fixed conditional parent-length signature.
- ϕ_{ref} : fixed, cross-fitted global reference inverse-dispersion.

- $z_{s,c}$: inferred effective reference-cell-equivalent, depth-scaled abundance.

The final cell-type composition is:

$$\pi_{s,c} = \frac{z_{s,c}}{\sum_{c'} z_{s,c'}}.$$

Main idea

ShapeMix-ATAC asks one controlled question: *does parent-fragment length composition improve deconvolution beyond peak totals?* The count-only arm uses the exact same total likelihood and optimization; ShapeMix adds only the conditional length-composition term.

References and positioning notes

References

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